An HF-ordering is over the set it closes

Silvia Barbina, Enrique Casanovas

arxiv:2411.13723v3; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's labelled environment at source0.tex lines 346-347 and its proof environment opens at line 348; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. This source is newly approved for this line; the preamble is built from the author's own package lines, macro block and theorem-environment declarations (source lines 4-65), with the author's optioned amsmath line omitted so that it cannot clash with the wrapper's amsmath.

Editorial note. The source carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label.

Editorial note. Metadata provenance: the source record of the pinned copy and the licence evidence file are both present; the licence is CC BY 4.0 with status supported.

Editorial note. r002 note: the pre-publication gate compares the card's source_version with the source_version recorded in the licence evidence of the same source. This source's evidence file records exactly "2411.13723v3", while the card had carried "arXiv:2411.13723v3 (CC BY 4.0)" copied from the source record; r002 aligns the card with the evidence and changes nothing else. The evidence file itself is not modified.

Editorial scope. One original result of the pinned author manuscript, counted once: an HF-ordering is over the set it closes (author label Feb_23_2024_7).

Allowed standard prerequisites (editorial).

- HF-ordering over a subset. An ordering is over A when every element outside A has at most one pair of smaller elements forming a block with it.. It is the property the lemma establishes.

- Closure of a subset. The closure of A under the ordering is the smallest set containing A and closed under forming blocks whose other members are already present and smaller.. The hypothesis is that A equals its own closure.

Lemma 0.1 If $<$ is an HF-ordering of B and $A = \mathrm{cl}_<(A) \subseteq B$, then $<$ is an HF-ordering of B over A.

Proof. If not, there is some $b \in B \setminus A$ and there are distinct b_1, b_2, b_3, b_4, such that $\{b, b_1, b_2\}$, $\{b, b_3, b_4\}$ are blocks and, for each i, either $b_i < b$ or $b_i \in A$. Without loss of generality, $b_2 < b_1$. We claim that $b_1 < b$. If not, then $b_1 \in A$ and $b < b_1$, and since $A = \mathrm{cl}_<(A)$, we get $b \in A$, a contradiction. Hence, $b > b_1 > b_2$. By a similar argument, $b > b_3, b_4$. But then $<$ is not an HF-ordering. □